Óptica II

Chengyu Jin

27 de abril de 2026

Índice general

Capítulo 1

Quantum Ideal Gas

1.1. Non-interacting Ising model

$H = -J \sum_i s_i$ where $s_i = \pm 1$ (spin)

We must to find:

Z
A
$U = \langle H \rangle$
$\langle M \rangle = \langle \sum s_i \rangle / N$

For interacting case

At T_c my particles are disordered. Below T_c, symmetry is broken (spontaneous magnetization).

$Z = (z_i)^N$

$z_i = \sum_{s_i = \pm 1} e^{-\beta H_i} = e^{\beta J} + e^{-\beta J} = 2\cosh(\beta J)$

$Z = (2\cosh(\beta J))^N$

$A = -kT \ln Z = -NkT \ln(2\cosh(\beta J))$

$U = \left(-\frac{\partial \ln Z}{\partial \beta}\right)_{N,V} = -\frac{\partial}{\partial \beta}(N \ln(2\cosh(\beta J))) = -N\frac{2\sinh(\beta J)}{2\cosh(\beta J)} \cdot J = -NJ\tanh(\beta J)$

In other way:

$U = \langle H \rangle = -J \sum_i \langle s_i \rangle = -NJ\tanh(\beta J)$
$\langle s_i \rangle = 1 \cdot P(s_i = 1) - 1 \cdot P(s_i = -1) = \frac{2\sinh(\beta J)}{2\cosh(\beta J)} = \tanh(\beta J)$

where $P(s_i) = \frac{e^{-\beta J s_i}}{2\cosh(\beta J)}$
$P(1) = \frac{e^{\beta J}}{2\cosh(\beta J)}$
$P(-1) = \frac{e^{-\beta J}}{2\cosh(\beta J)}$

1.2. Ideal Quantum Gas

System made of N identical particles:

$$\Psi(r_1, r_2, \ldots, r_N, t) \tag{1.1}$$

Its evolution is given by:

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi \quad ; \quad H = \sum_{i=1}^{N} \frac{p_i^2}{2m} + V(r_1, r_2, \ldots, r_N) \tag{1.2}$$

P_{ij} is the **permutation operator** which exchanges the labels of particles i and j:

$$P_{ij}\Psi(\ldots, r_i, \ldots, r_j, \ldots, t) = \Psi(\ldots, r_j, \ldots, r_i, \ldots, t) \tag{1.3}$$

$[H, P_{ij}] = 0 \quad \forall i, j \quad (i \neq j)$

If $H\phi_n = E_n\phi_n$:

$$\phi(r_1, \ldots, r_N, t) = \sum a_n \phi_n(r_1, \ldots, r_N) \quad ; \quad a_n = \langle \phi_n | \Psi \rangle \tag{1.4}$$

Now we apply P_{ij} to Ψ and then H:

$$\begin{aligned}
&HP_{ij}\Psi(\ldots, r_i, \ldots) \rightarrow \\
&H\Psi(\ldots, r_j, \ldots, r_i, \ldots, t) = \sum a_n E_n \phi_n(\ldots, r_j, \ldots, r_i, \ldots, t) = \sum a_n E_n \underbrace{P_{ij}}_{\text{scalars}} \phi_n(\ldots, r_i, \ldots, r_j, \ldots, t) \\
&= P_{ij} \sum a_n E_n \phi_n(\ldots, r_i, \ldots, r_j, \ldots, t) = P_{ij} H\Psi \\
&\rightarrow \text{We have proved that } [H, P_{ij}] = 0 \quad \forall i, j \quad (i \neq j)
\end{aligned}$$

Commuting observables have a complete set of common eigenfunctions:

$$P_{ij}H\phi = P_{ij}E\phi = E(P_{ij}\phi) \xrightarrow{[H, P_{ij}]=0} H(P_{ij}\phi) \tag{1.5}$$

This implies $P_{ij}\phi$ is also an eigenfunction of H with the eigenvalue E.

$P_{ij}\phi = \Theta\phi$

Value of Θ?

$$\begin{aligned}
P_{ij}\phi(\ldots, r_i, \ldots, r_j, \ldots, t) &= \phi(\ldots, r_j, \ldots, r_i, \ldots, t) \\
P_{ij}^2\phi(\ldots, r_i, \ldots, r_j, \ldots, t) &= \phi(\ldots, r_i, \ldots, r_j, \ldots, t) = I\phi
\end{aligned}$$

Then $P_{ij}^2 = I$.

As $P_{ij}\phi = \Theta\phi$:

$$P_{ij}\phi(\dots, r_i, \dots, r_j, \dots, t) = \Theta\phi(\dots, r_i, \dots, r_j, \dots, t)$$
$$P_{ij}^2\phi(\dots, r_i, \dots, r_j, \dots, t) = \Theta^2\phi(\dots, r_i, \dots, r_j, \dots, t)$$

As $P_{ij}^2 = I$, then $\Theta^2 = 1 \implies \Theta = \pm 1$.

- **Bosons:** $\Theta = +1 \rightarrow$ *Symmetric*
- **Fermions:** $\Theta = -1 \rightarrow$ *Antisymmetric*

What happens if two particles are in the same position ($r_i = r_j$) and I exchange them with P_{ij}?

First case (Fermions): $P_{ij}\phi(\dots, r_i, \dots, r_i, \dots, t) = -\phi(\dots, r_i, \dots, r_i, \dots, t)$ $\implies \phi(\dots, r_i, \dots, r_i, \dots, t) = 0 \quad \rightarrow$ It is **forbidden** that two particles occupy the same position. These particles are known as **Fermions** (half-integer spin).

Second case (Bosons): $P_{ij}\phi(\dots, r_i, \dots, r_i, \dots, t) = \phi(\dots, r_i, \dots, r_i, \dots, t)$ $\implies \phi(\dots, r_i, \dots, r_i, \dots, t) = \phi(\dots, r_i, \dots, r_i, \dots, t) \quad \rightarrow$ It is **not forbidden** that two particles occupy the same position. These particles are known as **Bosons** (integer spin).

Examples:

Fermions	**Bosons**
Electron	Quark
Proton	Gluon
Neutron	α particle
$\vdots$	$\vdots$
ϕ is antisymmetric ($\Theta = -1$)	ϕ is symmetric ($\Theta = +1$)

Ideal $\rightarrow H = \sum_i^N H_i$

$\sum_i n_i = N$
$\sum_i n_i \epsilon_i = E$

Fermions vs Bosons occupancy:

- **Fermions:** $n_j \in \{0, 1\}$ (Pauli exclusion principle).
- **Bosons:** $n_j \in \{0, 1, 2, \dots, N\}$ (No restriction).

$$Z(T,V) = \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i} = \sum_{\{n_i\}} e^{-\beta E(\{n_i\})} \delta_{N, \sum n_i} \tag{1.6}$$

$\rightarrow$ *This restriction ($\sum n_i = N$) does not allow us to perform the sum easily.*

Boltzmann approximation $\rightarrow z_N = \sum_{\{n_i\}} \frac{n_1! n_2! \ldots n_i!}{N!} e^{-\beta \sum n_i \epsilon_i} \approx \frac{1}{N!} \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i}$

Ensembles Comparison:

- **Microcanonical:** (E, V, N) fixed. Isolated system.
- **Canonical:** $(\langle E \rangle, V, N)$ fixed. Thermal contact with reservoir at T.
- **Grand Canonical (Macrocanonical):** $(\langle E \rangle, V, \langle N \rangle)$ fixed. Thermal and particle contact (μ, T).